

Assignment-3 : Chapter-3

Assignment Date: January 14th, 2026

Deadline: January 28th, 2026

Class: CSIT 5th Semester, New Summit College

Subject: Simulation & Modeling

Submission Mode: **Handwritten Document**

Questions:

1. What do you mean by a queuing system? Explain the basic characteristics of a queuing system with an example. **2079**
2. Define the queuing system. Explain the elements of a queuing system with an example.
3. Define congestion. What do you mean by server utilization?
4. What do you mean by multi-server queues?
5. Define queuing discipline. Describe different types of queuing disciplines with examples. Also, explain different performance measures for the evaluation of a queuing system. **2076**
6. What are the various performance measures in a single-server queuing System? Explain which of them determines system stability and how? **2078**
7. Write Short Notes on: **2081**
 - a. Queuing Discipline
 - b.
8. Define traffic intensity and server utilization. Explain Kendall's notation for a queuing system with an example. **2081**
9. What is the calling population? Explain the arrival and service process in a queue. **2080**
10. Kendall Notation Questions:
 - a. Explain the following Kendall's Notation in detail:
 - i. M/M/1
 - ii. M/M/3
 - iii. M/N/2/10
 - iv. M/M/4/20

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v. D/M/5

b. Derive the Kendall's Notation for the following questions:

- i. A hospital emergency room has 6 doctors treating patients. Patient arrivals follow a general distribution, and service times also follow a general distribution. There is no system capacity limit, and patients are served in priority order. Express this system in Kendall notation.
- ii. A computer network has a server cluster with 10 parallel processors. Jobs arrive according to a Poisson process, and service times follow an exponential distribution. The system has a total capacity of 100 jobs, and they are processed based on a shortest-job-first discipline. What is the correct Kendall notation?
- iii. A barber shop has 3 barbers, each serving one customer at a time. The arrivals follow a Poisson process, and service times follow an exponential distribution. However, if a customer sees more than 5 people waiting, they leave without getting a haircut (balking). Express this system in Kendall notation.
- iv. A manufacturing assembly line has a conveyor system with 4 robotic arms handling parts. Parts arrive based on a deterministic schedule, and each robotic arm takes a normally distributed time to process a part. The system can hold up to 50 parts at a time, and processing follows a FIFO (First-In-First-Out) discipline. What is the Kendall notation?
- v. A cloud computing server has an infinite calling population, with arrivals following a general distribution. The service times are also general, and there are 8 parallel servers available. The system has a total capacity of 200 tasks, and the queue discipline follows round-robin scheduling. How would you express this in Kendall notation?